Dr Andrew R. **McCluskey**

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An experienced research leader with an international reputation for fundamental research working at the interface between the physical sciences, statistics and applied mathematics to improve neutron scattering and computational simulation methods. Educator and developer of open educational resources, passionate about training physical scientists in data skills and engaging others in scientific research and practice. Driven by a collaborative approach to research, a commitment to open science, and a passion for excellent mentorship.

Employment

University of Bristol — *BRISTOL, GB*

Senior Lecturer (UK-Equivalent of Associate Professor) in Chemistry — *FEB. 2026 - ONGOING*

Lecturer in Chemistry — *AUG. 2023 - JAN. 2026*

European Spallation Source ERIC — *KØBENHAVN, DK & LUND, SE*

Instrument Data Scientist for Neutron Reflectometry — *JAN. 2021 - AUG. 2023*

Diamond Light Source — *HARWELL-OXFORD, GB*

Data Analysis Scientist – Reflectivity — *APR. 2019 - DEC. 2020*

Education

University of Bristol — *BRISTOL, GB*

Postgraduate Certificate in Academic Practice — *JAN. 2024 - JUL. 2025*

University of Bath & Diamond Light Source — *BATH & HARWELL-OXFORD, GB*

PhD in Chemistry — *SEPT. 2015 - APR. 2019*

University of Edinburgh — *EDINBURGH, GB*

MChem in Materials Chemistry with a Year in Industry – First Class — *SEPT. 2010 - JUN. 2015*

Invited Positions

Diamond Light Source — *HARWELL-DIDCOT, GB*

Visiting Scientist — *MAR. 2024 - ONGOING*

European Spallation Source ERIC — *KØBENHAVN, DK & LUND, SE*

Visiting Scientist — *SEPT. 2023 - SEPT. 2024*

University of Bath — *BATH, GB*

Visiting Lecturer — *SEPT. 2019 - SEPT. 2022*

ISIS Neutron Training Course — *HARWELL-OXFORD, GB*

Lecturer — *MAR. 2017 - MAR. 2018*

Funding Secured

To Date **Awarded 25 days of competitive neutron scattering beamtime across a range of instrumentation**, National and International Neutron Scattering Facilities — *approx. £500 000* (Lead Investigator)

2025/12 **Digital engineering biology - Accelerating digital biology from models to innovation**, University of Bristol, Faculty of Science and Engineering Strategic Research Accelerator — *£134 000* (Co-investigator)

2025/06 **Support for the Solid Data Summer School**, Royal Society of Chemistry Solid State Chemistry Group — *£10 000* (Co-organiser)

2024/08 **A unique cutting-edge Atomic Force Microscopy platform to converge frontier biosciences with engineering, environmental and physical sciences**, Biotechnology and Biological Sciences Research Council (BBSRC) ALERT — *£1 000 000* (Co-investigator)

2023/07 **Conference grant for DMSC Summer School**, Carlsberg Fondet — *60 000 DKK* (Project lead)

2023/07 **Sponsorship of DMSC Summer School**, DanScat — *25 000 DKK* (Project lead)

2020/08 **Development of a Bayesian regularisation framework for the analysis of reflectometry**, Diamond Light Source Year in Industry Studentship 2020/2021 — *approx. £20 000* (Project lead)

2019/11 **pythoninchemistry Hackathon**, Royal Society of Chemistry Higher Education Group Kickstart Scheme — *£300* (Co-investigator)

Awards/Memberships

2025/11/05 **Springboard Award**, ISIS Neutron and Muon Source

2025/07/25 **Fellow of the Higher Education Academy (FHEA)**, Advance HE

2024/04/11 **B.T.M. Willis Prize**, UK Neutron Scattering Group

2023/12/19 **Materials Advances Computational Science Prize**, Royal Society of Chemistry Solid State Chemistry Group Christmas Meeting

2018/10/12 **IUCr Journals Prize for the Best Student Lecture**, SAS2018

2018/06/14 **The Computational Prize – Best Oral Presentation**, University of Bath Bolland Symposium

2018/05/17 **Nominated for Faculty Teaching Assistant Award**, University of Bath Faculty of Science

2017/06/12 **Best Talk Award – Sponsored by Santander**, University of Bath Faculty of Science Graduate School Research Afternoon

Service/Community

Royal Society of Chemistry Material Chemistry Community Committee

Appointed Member — *JUN 2026 - PRESENT*

Journal of Open Source Education

Topic Editor (Computational Chemistry & Physics) — *OCT 2022 - PRESENT*

Open Reflectivity Standards Organisation

Founding Member — *OCT. 2019 - JUN. 2025*

RSC/IOP Neutron Scattering Group Committee

Treasurer — *OCT. 2024 - PRESENT*

Ordinary Committee Member — *JUN. 2023 - SEPT. 2024*

Early Career Representative — *JUN. 2017 - JUN. 2023*

Royal Society of Chemistry

Member — *SEPT. 2010 - PRESENT*

Language Fluency

Mother tongue English

CEFR-level B2 Danish (Prøve i Dansk 3)

Publications

ORCiD: 0000-0003-3381-5911.

28. T. Zolzaya & **A. R. McCluskey*** Numerical and analytical strategies to propagate correlated uncertainties through the discrete Fourier transform *Submitted,* 2026. Available on arXiv: 10.26434/chemrxiv.15008407.
27. J. Dunn*, **A. R. McCluskey**, A. S. F. Oliveira*, N. L. Allan* & A. J. Mulholland*. An Efficient Temperature-Jump Method for the Calculation of Heat Capacities, *Submitted,* 2026. Available on ChemRxiv: 10.26434/chemrxiv.15008407.
26. H. Richardson, K. McColl, G. Nilsen, J. Armstrong*, & **A. R. McCluskey***. Lost in Translation: Simulation-Informed Bayesian Inference Improves Characterization of Molecular Motion From Neutron Scattering, *J. Phys. Chem. Lett.,* **17**(29), 8411-8418, 2026. DOI: 10.1021/acs.jpclett.6c01399.
25. B. J. Morgan* & **A. R. McCluskey***. Uncertainty in MD-Derived Diffusion Coefficients Depends on Analysis Protocol, Not Just Simulation Data, *Solid State Ionics,* **444**, 117249 2026. DOI: 10.1016/j.ssi.2026.117249.
24. **A. R. McCluskey***, S. W. Coles, & B. J. Morgan*. Bayesian Methods for the Investigation of Temperature-Dependence in Conductivity, *PRX Energy,* **5**, 022001, 2026. DOI: 10.1103/nxgq-lmp6.
23. B. A. Humphreys*, P. Gutfreund, **A. R. McCluskey**, T. Arnold, J. Vind, & T. Nylander. Following the structural changes of triolein films during lipolysis, *Soft Matter,* **22**(2), 343-354, 2026. DOI: 10.1039/D5SM00820D.
22. H. Richardson, J. Dunn*, & **A. R. McCluskey***. On the Estimation of Centre of Mass in Periodic Systems, *J. Chem. Phys.,* **162**(20), 204103, 2025. DOI: 10.1063/5.0260928.
21. **A. R. McCluskey***, S. W. Coles, & B J. Morgan*. Accurate Estimation of Diffusion Coefficients and their Uncertainties from Computer Simulation, *J. Chem. Theory Comput.,* **21**(1), 79-87, 2025. DOI: 10.1021/acs.jctc.4c01249.
20. N. Shiaelis*, L. A. Clifton, & **A. R. McCluskey***. Investigating model influence on the analytical resolution of neutron reflectometry, *Preprint,* 2024. Available on arXiv: 2403.13566.
19. **A. R. McCluskey***, M. Rivera*, & A. S. J. S. Mey*. Digital Skills in Chemical Education, *Nat. Chem.,* **16**(9), 1383-1384, 2024. DOI: 10.1038/s41557-024-01613-x.

18. J. Dunn, J. M. Crossley-Lewis, **A. R. McCluskey**, F. Jackson, C. Buda, G. Sunley, A. J. Mulholland, & N. L. Allan.* Diffusion Mechanisms and Preferential Dynamics of Promoter Molecules in ZSM-5 Zeolite, *Catal. Sci. Technol.,* **14**(13), 3674-3681, 2024. DOI: 10.1039/D4CY00506F.
17. **A. R. McCluskey***, P. Aulin, F. Bolmsten, M. Bertelsen, C. M. C. Lobley, J. Lewis, M. Novelli, C. Somani, A. Stefanov, M. Trajanovski, N. Vaytet, P. K. Willendrup, J.-L. Wynen, S. Yoo, & T. H. Rod*. The First European Spallation Source Data Management and Software Centre Summer School, *Neutron News,* **35**(1), 1-3, 2024. DOI: 10.1080/10448632.2024.2331387.
16. **A. R. McCluskey***, A. G. Squires, J. Dunn, S. W. Coles, & B. J. Morgan*. kinisi: Bayesian analysis of mass transport from molecular dynamics simulations, *J. Open Source Softw.,* **9**(94), 5984, 2024. DOI: 10.21105/joss.05984.
15. **A. R. McCluskey***. Is there still a place for linearization in the chemistry curriculum?, *J. Chem. Educ.,* **100**(11), 4174-4176, 2023. DOI: 10.1021/acs.jchemed.3c00466.
14. G. Krenzer, J. Klarbring, K. Tolborg, H. Rossignol, **A. R. McCluskey**, B. J. Morgan*, & A. Walsh*. Nature of the Superionic Transition of Lithium Nitride from Machine Learning Force Fields, *Chem. Mater.,* **35**(15), 6133-6140, 2023. DOI: 10.1021/acs.chemmater.3c01271.
13. **A. R. McCluskey***, A. J. Caruana*, C. J. Kinane, A. J. Armstrong, T. Arnold, J. F. K. Cooper, D. L. Cortie, A. V. Hughes, J.-F. Moulin, A. R. J. Nelson, W. Potrzebowski, & V. Starostin. Advice on describing Bayesian analysis of neutron and X-ray reflectometry, *J. Appl. Crystallogr.,* **56**(1), 12-17, 2023. DOI: 10.1107/S1600576722011426.
12. R. Brearton*, **A. R. McCluskey**, & T. Snow. islatu: A Python package for the reduction of reflectometry data, *J. Open Source Softw.,* **7**(77), 4397, 2022. DOI: 10.21105/joss.04397.
11. T. Arnold*, A. Terry, E. Blackburn, U. Hejral, Z. Heyles, **A. R. McCluskey**, T. Nylander, & M. Wolff. The 16th International Conference on Surface X-ray and Neutron Scattering (SXNS16), *Neutron News,* **33**(2), 2, 2022. DOI: 10.1080/10448632.2022.2050633.
10. T. Arnold*, B. Murphy, **A. R. McCluskey**, J. Stahn, & M. W. A. Skoda. A Report on the Third Meeting of the Open Reflectivity Standards Organisation (ORSO), *Neuton News,* **33**(1), 2, 2022. DOI: 10.1080/10448632.2021.2005422.
9. J. M. Dean, S. W. Coles*, W. R. Saunders, **A. R. McCluskey**, M. J. Wolf, A. B. Walker, & B. J. Morgan*. Overscreening and Underscreening in Solid-Electrolyte Grain Boundary Space-Charge Layers, *Phys. Rev. Lett.,* **127**(13), 135502, 2021. DOI: 10.1103/PhysRevLett.127.135502.
8. A. Markvardsen*, T. Rees, M. Wathen, A. Lister, P. Odagiu, A. Anuchitanukul, T. Farmer, A. Lim, F. Montesino, T. Snow, & **A. McCluskey**. FitBenchmarking: an open source Python package comparing data fitting software, *J. Open Source Softw.,* **6**(62), 3127, 2021. DOI: 10.21105/joss.03127.
7. **A. R. McCluskey**, K. S. W. Hung, B. Marzec, J. O. Sindt, N. A. J. M. Sommerdijk, P. J. Camp, & F. Nudelman*. Disordered Filaments Mediate the Fibrillogenesis of Type-I Collagen in Solution, *Biomacromolecules,* **21**(9), 3631-3643, 2020. DOI: 10.1021/acs.biomac.0c00667.
6. **A. R. McCluskey***, J. F. K. Cooper, T. Arnold, & T. Snow. A general approach to maximise information density in neutron reflectometry analysis, *Mach. Learn.: Sci. Technol.,* **1**(3), 035002, 2020. DOI: 10.1088/2632-2153/ab94c4.
5. **A. R. McCluskey***, & T. Snow. uravu: making Bayesian data analysis easy(er), *J. Open Source Softw.,* **5**(50), 2214, 2020. DOI: 10.21105/joss.02214.
4. **A. R. McCluskey***, J. Grant, A. J. Smith, J. L. Rawle, D. J. Barlow, M. J. Lawrence, S. C. Parker, & K. J. Edler*. Assessing molecular simulation for the analysis of lipid monolayer reflectometry, *J. Phys. Comm.,* **3**(7), 075001, 2019. DOI: 10.1088/2399-6528/ab12a9.
3. **A. R. McCluskey***, J. Grant, A. R. Symington, T. Snow, J. Doutch, B. J. Morgan*, S. C. Parker, & K. J. Edler. An introduction to classical molecular dynamics simulation for experimental scattering users, *J. Appl. Crystallogr.,* **52**(3), 665-668, 2019. DOI: 10.1107/S1600576719004333.
2. **A. R. McCluskey***, A. Sanchez-Fernandez, K. J. Edler, S. C. Parker, A. J. Jackson, R. A. Campbell, & T. Arnold*. Bayesian determination of the effect of a deep eutectic solvent on the structure of lipid monolayers, *Phys. Chem. Chem. Phys.,* **21**(11), 6133-6141, 2019. DOI: 10.1039/C9CP00203K.
1. **A. R. McCluskey***, B. J. Morgan, K. J. Edler, & S. C. Parker. pylj: A teaching tool for classical atomistic simulation, *J. Open Source Educ.,* **1**(2), 19-21, 2018. DOI: 10.21105/jose.00019.
0. **A. R. McCluskey**, & K. J. Edler*. Model-dependent Small-angle Scattering for the Study of Complex Organic Materials, *Curr. Org. Chem.,* **22**(8), 750-757, 2018. DOI: 10.2174/1875692115666170612104439.

* Denotes corresponding authorship.

Presentations

Invited Talks

2026/07/27 **Methods Matter: A Case for Careful Analysis of Molecular Simulation**, CCP5 Summer School 2026 *@ NEWCASTLE, GB*

2026/04/02 **Bayesian Model Selection Provides a Robust Tool for Experimental Analysis**, Machine Learning for Experimental Materials Data *@ LONDON, GB*

2026/01/16 **A Case for Good Simulation To Complement Neutron Scattering: A p-QENS Example**, Neutron Scattering Group Early Career Meeting *@ BRISTOL, GB*

2026/01/15 **Accurate Diffusion Estimation From Simulation and the Impacts on Temperature Dependent Modelling**, University of Bristol, School of Mathematics, Fluids and Materials Seminar *@ BRISTOL, GB*

2025/06/03 **Accurate Estimation of Self- and Collective-Motion from Molecular Dynamics Simulation**, PAQMAN *@ STAVANGER, NO*

2025/05/01 **Accurate Diffusion Estimation From Simulation and the Impacts on Temperature Dependent Modelling**, DAMTP BioLunch *@ CAMBRIDGE, GB*

2025/04/16 **Accurate Diffusion Estimation From Simulation**, Digital Chemistry Festival – **Plenary Speaker** *@ BRISTOL, GB*

2025/02/20 **Accurate Diffusion Estimation From Simulation: Can We Improve QENS Analysis?**, ISIS Research Seminar *@ HARWELL, GB*

2025/01/19 **Accurate Diffusion Estimation From Simulation**, Bragg Exchange 2025 *@ LEEDS, GB*

2024/11/27 **Accurate Diffusion Estimation From Simulation and Propagation to Temperature Dependent Modelling**, University of Warwick Computational Chemistry Seminar *@ WARWICK, GB*

2024/06/26 **A Short History of Trying To Make Computational Research More Reproducible**, Reproducibility By Design Workshop *@ BRISTOL, GB*

2024/06/05 **Accurate Diffusion Estimation From Simulation and Propagation to Temperature Dependent Modelling**, University of Birmingham Research Seminar *@ BIRMINGHAM, GB*

2024/04/11 **How To Care About Data in Neutron Scattering: A Biased Perspective**, UK Neutron and Muon Science User Meeting 2024 – **B.T.M. Willis Prize Lecture** *@ WARWICK, GB*

2024/03/12 **Introduction to Reflectometry: With a Focus on Analysis**, French-Swedish Academy for Scattering Experiments and Modelling for Life Science *@ LUND, SE*

2022/07/05 **Developing and sharing an undergraduate chemistry course for Python**, 2022 RACI National Congress, Physical and Computational Chemistry Education Symposium – **Keynote Speaker** *@ HYBRID (BRISBANE, AU)*

2020/04/03 **Reflectometry and data science**, #theLightStuff Webinars (youtu.be/PHBLK_3sfi8) *@ ONLINE*

2017/06/19 **Surfactants and Molecular Dynamics**, CCP-SAS Joint Meeting, Cardiff University *@ CARDIFF, GB*

2017/06/12 **Putting computers to work for large experiments**, Faculty of Science Graduate School Research Afternoon, Bath University – **Best Talk Award** *@ BATH, GB*

2016/05/23 **SAS, Sims and Soft Matter Self-Assembly**, CCP-SAS Joint Meeting, NIST *@ GAITHERSBURG, US*

Contributed Talks

2026/04/14 **Lost in Translation: Simulation-Informed Bayesian Inference for Quasi-Elastic Neutron Scattering**, UK Neutron and Muon Science User Meeting 2026 *@ WARWICK, GB*

2024/07/03 **Accurate Diffusion Estimation From Simulation and Propagation to Temperature Dependent Modelling**, Workshop on Synergies Between Mathematics, Data Science, and Molecular Simulations in Materials Science *@ BIRMINGHAM, GB*

2023/09/22 **A perspective on probabilistic chemical models to complement quantum algorithms**, Scientific Applications of Quantum Computing: Materials, Chemistry and Biology *@ LONDON, GB*

2022/09/21 **PaN-Training e-Learning: education and training for scientists and students**, NOBUGS (New Opportunities for Better User Group Software) 2022 *@ HYBRID (VILLIGEN, CH)*

2022/08/22 **Using Bayesian inference as a tool to more completely understand neutron reflectometry**, International Conference on Neutron Scattering 2022 *@ BUENOS AIRES, AR*

2019/07/12 **Automating reflectometry reduction and analysis at Diamond Light Source**, M4 COLLOIDS *@ BATH, GB*

2019/05/07 **Bayesian determination of the effect of a deep eutectic solvent on the structure of lipid monolayers**, BAYES@LUND 2019 *@ LUND, SE*

2018/10/30 **Comparing coarse-grained simulation-derived and traditional analysis method for monolayer reflectometry**, Trends and Perspectives in Neutron Instrumentation *@ TUTZING, DE*

2018/10/12 **Using high-performance computing and molecular dynamics to rationalise micelle structure from small-angle scattering**, SAS2018 *@ TRAVERSE CITY, US*

2018/10/09 **pylj: an open-source Python library for teaching the interaction between molecular simulation and scattering**, SAS2018 – **Best Student Lecture Prize** *@ TRAVERSE CITY, US*

2018/09/16 **Introducing programming to undergraduate chemists: and the tools we've developed to help them**, PYCON GB *@ CARDIFF, GB*

2018/08/23 **Introducing programming to undergraduate chemists: and the tools we've developed to help them**, VICEPHEC18 *@ SHEFFIELD, GB*

2018/06/14 **Using markov chain monte-carlo to estimate uncertainties in x-ray reflectometry modelling**, University of Bath Bolland Symposium *@ BATH, GB*

2018/02/09 **Probabilistic analysis of reflectometry data: Phospholipids at the DES-air interface**, Neutrons and Global Challenges II: Health and Healthcare *@ LONDON, GB*

2017/09/12 **Simulations to understand reflectivity: How coarse can we go?**, CCP5 AGM *@ GLASGOW, GB*

2017/04/13 **Simulations to understand reflectivity: How coarse can we go?**, Faraday Joint Interest Group Conference *@ WARWICK, GB*

2017/03/23 **Coarse graining and reflectivity: a love story?**, CompChem Seminar, University of Bath *@ BATH, GB*

2017/02/28 **Reflectivity: from simulation to experiment**, International Soft Matter Workshop *@ HELFORD, GB*

2016/06/23 **Smart analysis of soft matter**, Bombannes Summer School *@ BOMBANNES, FR*